

Companion LXIV: Wasserstein Distances of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Wasserstein distances provide a metric-space comparison of persistence diagrams, quantifying the difference between topological structures across filtrations. In weak-lensing analyses, these distances measure how the topology of convergence fields changes with redshift, smoothing scale, or cosmological model. In the standard cosmological model, Wasserstein distances reflect the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the metric structure of persistence diagrams in a characteristic way. This Companion develops a continuous description of Wasserstein distances in RDCC, deriving how the relaxation scale influences diagram geometry, topological stability, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive metric-space probe of the RDCC gravitational field.

1 Introduction

Persistent homology provides a topological description of cosmological fields by tracking the birth and death of connected components and loops across a filtration. Wasserstein distances extend this by defining a metric on the space of persistence diagrams, allowing a quantitative comparison of topological structure.

In the standard cosmological model, Wasserstein distances reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the metric structure of persistence diagrams in a scale-dependent manner, producing a distinctive signature in Wasserstein-distance analyses.

2 Persistence Diagrams and Wasserstein Distances

A persistence diagram D consists of points (b, d) representing the birth and death of topological features. The p -Wasserstein distance between two diagrams D_1 and D_2 is

$$W_p(D_1, D_2) = \left[\inf_{\gamma} \sum_{x \in D_1} \|x - \gamma(x)\|_{\infty}^p \right]^{1/p}, \quad (1)$$

where γ ranges over all bijections between the diagrams (including the diagonal).

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^{\alpha}], \quad (3)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^{\alpha}]. \quad (4)$$

The RDCC persistence diagram becomes

$$D_{\text{RDCC}} = \{(b_{\text{RDCC}}, d_{\text{RDCC}})\}. \quad (5)$$

4 Wasserstein Distances in RDCC

The Wasserstein distance between a standard and an RDCC diagram is

$$W_p(D, D_{\text{RDCC}}) = \left[\sum_i \|(b_i, d_i) - (b_{i,\text{RDCC}}, d_{i,\text{RDCC}})\|_{\infty}^p \right]^{1/p}. \quad (6)$$

This becomes

$$W_p(D, D_{\text{RDCC}}) = W_p(D, D) \left[1 - \chi_W \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (7)$$

reflecting the relaxation-driven modification of topological distances.

5 Metric-Space Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Wasserstein distances therefore exhibit:

- reduced distances between diagrams at large scales,
- enhanced distances at small scales,
- a characteristic turnover near ℓ_{rel} ,
- modified scaling with smoothing scale and redshift,
- altered clustering of diagrams in metric space.

These signatures provide a direct observational probe of the RDCC gravitational field in metric space.

6 Conclusion

Wasserstein distances of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the metric structure of persistence diagrams in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological metric spaces, nonlinear structure, and the interplay between light and gravity in RDCC.

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